

A Post-Quantum Crypto Agility Framework: Runtime Algorithm Negotiation, Quantum Threat Scoring, and Hybrid Key Exchange

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Abstract—The imminent arrival of cryptographically relevant quantum computers (CRQCs) demands that deployed systems migrate away from RSA and elliptic-curve cryptography toward post-quantum alternatives standardized by NIST. However, enterprise environments encompass thousands of heterogeneous cryptographic assets whose simultaneous migration is infeasible, creating a protracted transition period during which classical, hybrid, and post-quantum algorithms must coexist. We present QBRIDGE, an integrated framework addressing three critical facets of this transition. First, we introduce a *Crypto Agility Negotiation Protocol* that enables runtime algorithm selection between peers through capability advertisement, policy-constrained intersection, and zero-downtime rotation when algorithms are deprecated or compromised. The protocol supports 28 algorithms across six families governed by four organizational policies aligned with NIST SP 800-227, CNSA 2.0, and ETSI TS 103744. Second, we describe a *Quantum Threat Scoring Engine* that computes a composite Quantum Risk Score (QRS) in $[0, 100]$ by weighting five factors—algorithm vulnerability, time-horizon risk via Mosca’s inequality, data sensitivity, migration gap, and exposure surface—to prioritize migration across an entire cryptographic asset portfolio. Third, we specify three *Hybrid Key Exchange* constructions (X25519-ML-KEM-768, P384-ML-KEM-1024, X25519-ML-KEM-512) that combine classical ECDH with Module-LWE-based key encapsulation, deriving the shared secret through HKDF-SHA256 with domain separation, suitable for TLS 1.3 integration. We evaluate QBRIDGE on realistic workloads, demonstrating negotiation latency under 2 ms, scoring throughput exceeding 50 000 assets per second, and hybrid KEM overhead within 15% of standalone ML-KEM, making the framework practical for large-scale deployment.

Index Terms—post-quantum cryptography, crypto agility, hybrid key exchange, ML-KEM, quantum risk assessment, TLS 1.3, NIST PQC standards

I. INTRODUCTION

Shor’s algorithm [?] renders RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography insecure against a sufficiently large quantum computer. Although the exact timeline for a cryptographically relevant quantum computer (CRQC) remains uncertain, conservative estimates place this event within the 2030–2040 window [?]. The *harvest-now, decrypt-later* (HNDL) threat model makes the situation more acute: adversaries can record ciphertexts today and decrypt them once a CRQC becomes available [?]. For data with long confidentiality requirements—financial records, medical data,

classified communications—migration to quantum-resistant algorithms must begin *now*.

NIST finalized its first post-quantum standards in 2024: FIPS 203 (ML-KEM) [?] for key encapsulation, FIPS 204 (ML-DSA) [?] for digital signatures, and FIPS 205 (SLH-DSA) [?] for stateless hash-based signatures. Supplementary guidance in SP 800-227 [?] provides a migration roadmap, while the NSA’s CNSA 2.0 suite [?] imposes stricter timelines for national security systems. Despite these standards, the practical challenge of migrating a large enterprise from legacy cryptography to PQC remains daunting.

Three obstacles stand out. (i) *Heterogeneity*. A typical enterprise deploys dozens of distinct cryptographic algorithms across TLS termination, code signing, database encryption, VPN tunnels, and API authentication. No single algorithm or policy applies uniformly. (ii) *Risk prioritization*. With limited migration resources, organizations must decide *which* assets to migrate first. Intuitive approaches (“migrate everything using RSA”) ignore data sensitivity, exposure surface, and the HNDL timeline. (iii) *Transition coexistence*. During the multi-year migration window, hybrid constructions that combine classical and PQC algorithms are necessary both for defense-in-depth and for interoperability with peers that have not yet migrated [?].

A. Contributions

We present QBRIDGE, an integrated crypto agility framework that addresses all three obstacles through the following contributions:

- 1) **Crypto Agility Negotiation Protocol**. A runtime algorithm negotiation mechanism that advertises cryptographic capabilities, computes policy-constrained intersections, and supports zero-downtime algorithm rotation (Section III). The protocol manages 28 algorithms across six families under four configurable policies.
- 2) **Quantum Threat Scoring Engine**. A composite risk model that assigns a Quantum Risk Score (QRS) to every cryptographic asset in an organization’s portfolio, enabling quantitative migration prioritization based on Mosca’s inequality and five weighted factors (Section IV).
- 3) **Hybrid Key Exchange Constructions**. Three hybrid KEM variants for TLS 1.3—X25519-ML-KEM-768, P384-ML-

KEM-1024, and X25519-ML-KEM-512—with HKDF-based shared-secret combination and domain separation (Section V).

- 4) **Implementation and Evaluation.** A production-quality Python implementation with performance benchmarks demonstrating practical deployment characteristics (Section VI).

The remainder of the paper is organized as follows. Section II covers background and preliminaries. Sections III–V detail our three constructions. Section VI presents implementation results. Section VII provides a security analysis. Section VIII surveys related work, and Section IX concludes.

II. BACKGROUND AND PRELIMINARIES

A. Lattice-Based Cryptography

The NIST PQC standards are built primarily on the hardness of the *Module Learning with Errors* (Module-LWE) and *Module Short Integer Solution* (Module-SIS) problems [?].

[Module-LWE] Given a rank- k module over the polynomial ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, the Module-LWE problem asks to distinguish $(A, A \cdot s + e)$ from (A, u) where $A \xleftarrow{\$} R_q^{k \times k}$, $s \leftarrow \chi_s^k$, $e \leftarrow \chi_e^k$, and $u \xleftarrow{\$} R_q^k$, with χ_s, χ_e being centered binomial distributions with small parameter η .

ML-KEM (FIPS 203) instantiates an IND-CCA2 key encapsulation mechanism from Module-LWE via the Fujisaki–Okamoto transform. Three parameter sets—ML-KEM-512 (NIST Level 1), ML-KEM-768 (Level 3), and ML-KEM-1024 (Level 5)—provide increasing security margins at the cost of larger keys and ciphertexts [?].

ML-DSA (FIPS 204) is a digital signature scheme based on Module-LWE/Module-SIS with parameter sets ML-DSA-44, ML-DSA-65, and ML-DSA-87 [?]. SLH-DSA (FIPS 205) provides a conservative alternative based solely on hash-function security [?].

B. Quantum Threat Model

We adopt the standard *harvest-now, decrypt-later* (HNDL) threat model [?]. An adversary with classical computational power records ciphertexts and authenticated protocol transcripts today. At a future time t_{CRQC} , the adversary gains access to a CRQC capable of running Shor’s algorithm [?] and recovers plaintext from all recorded sessions using vulnerable key exchange algorithms.

Symmetric algorithms are affected by Grover’s algorithm [?], which provides a quadratic speedup for brute-force search, effectively halving the security level (e.g., AES-128 provides only 64-bit security against a quantum adversary, while AES-256 retains 128-bit security).

C. Mosca’s Inequality

Mosca’s theorem [?] provides a decision criterion for migration urgency:

$$x + y > z \implies \text{migration is overdue} \quad (1)$$

where x is the number of years the data must remain confidential, y is the time (in years) required to complete migration, and z is the estimated time until a CRQC can break the algorithm. When $x + y > z$, data encrypted today will still require protection after the quantum threat materializes, and the migration cannot complete in time.

D. Standards Landscape

Our framework is aligned with four major standards:

- **NIST SP 800-227** [?]: Provides a phased transition roadmap for federal agencies to adopt PQC.
- **CNSA 2.0** [?]: The NSA’s algorithm suite mandating ML-KEM-1024, ML-DSA-87, and AES-256 for national security systems by 2035.
- **ETSI TS 103 744** [?]: European migration strategies including hybrid approaches.
- **RFC 8446** [?]: TLS 1.3 cipher suite negotiation, which our agility protocol extends for PQC.

III. CRYPTO AGILITY NEGOTIATION PROTOCOL

Crypto agility—the ability to switch algorithms without redesigning protocols—has been identified as a critical requirement for the PQC transition [?], [?], [?]. We formalize a negotiation protocol that supports runtime algorithm selection, policy enforcement, and zero-downtime rotation.

A. Design Goals

Our protocol satisfies five design goals:

- G1. Forward Compatibility:** Unknown future algorithms can be negotiated via OID or canonical name without protocol changes.
- G2. Hybrid Negotiation:** Classical and PQC algorithms operate simultaneously during the transition period.
- G3. Policy Enforcement:** Organizational policies constrain negotiation to approved algorithms with minimum security levels.
- G4. Zero-Downtime Rotation:** Algorithms can be banned or upgraded without interrupting active sessions.
- G5. Audit Trail:** Complete negotiation history is maintained for compliance reporting.

B. Algorithm Registry

The protocol maintains a registry of 28 algorithms organized into six families: KEM, signature, symmetric, hash, MAC, and KDF. Table I summarizes the key entries.

Each algorithm is assigned a lifecycle status from the ordered set {RECOMMENDED, ACCEPTABLE, LEGACY, DEPRECATED, BANNED}, and a security era from {CLASSICAL, HYBRID, POST-QUANTUM, CNSA-2.0}.

C. Organizational Policies

Four pre-defined policies govern negotiation behavior:

Transitional: ≥ 128 -bit security, hybrid required, classical-only disallowed. Suitable for organizations beginning PQC migration.

TABLE I
SELECTED ENTRIES FROM THE ALGORITHM REGISTRY.

Algorithm	Family	Bits	Level	PK (B)	CT/Sig (B)
ML-KEM-512	KEM	128	1	800	768
ML-KEM-768	KEM	192	3	1 184	1 088
ML-KEM-1024	KEM	256	5	1 568	1 568
X25519-ML-KEM-768	KEM	192	3	1 216	1 120
P384-ML-KEM-1024	KEM	256	5	1 665	1 665
ML-DSA-44	Sig	128	2	1 312	2 420
ML-DSA-65	Sig	192	3	1 952	3 293
ML-DSA-87	Sig	256	5	2 592	4 595
Falcon-512	Sig	128	1	897	666
SLH-DSA-128f	Sig	128	1	32	17 088
AES-256-GCM	Sym	256	–	32	–
SHA3-256	Hash	256	–	–	32
HKDF-SHA3-256	KDF	256	–	–	–

Post-Quantum: ≥ 192 -bit security, NIST Level ≥ 3 , classical algorithms banned. For environments ready for full PQC deployment.

CNSA 2.0: ≥ 256 -bit security, NIST Level 5, prefers ML-KEM-1024 and ML-DSA-87. Targets national security compliance.

Legacy Compatible: ≥ 128 -bit security, classical-only permitted. For backward compatibility with legacy peers.

D. Negotiation Protocol

The negotiation proceeds in three phases, formalized in Algorithm 1.

The DETERMINEERA function classifies the negotiated suite: if all selected algorithms belong to the post-quantum or CNSA-2.0 era, the suite era is set to POST-QUANTUM; if at least one algorithm is post-quantum while others are classical, the era is HYBRID; otherwise the era is CLASSICAL.

E. Zero-Downtime Rotation

When a vulnerability is discovered in an active algorithm, the framework supports in-place renegotiation without session teardown:

The rotation is atomic with respect to the session: the current suite remains active until a new suite is successfully negotiated, ensuring no interruption in cryptographic protection.

F. Compliance Auditing

Every negotiation event is appended to an immutable history with the following fields: suite identifier, selected algorithms, security era, initiator and responder identities, timestamp, and validity period. This history supports compliance reporting against NIST SP 800-227 and CNSA 2.0 timelines.

IV. QUANTUM THREAT SCORING ENGINE

Risk-based prioritization is essential when migration resources are limited [?]. We formalize a scoring model that assigns each cryptographic asset a Quantum Risk Score (QRS), enabling quantitative comparison across an organization's portfolio.

Algorithm 1 Crypto Agility Negotiation

Input: Local policy π , local capability C_L , remote capability

C_R

Output: Negotiated suite $\mathcal{S}$ or $\perp$

```

1: Phase 1: Capability Advertisement
2:  $C_L \leftarrow \text{FILTERBYPOLICY}(\pi, \text{Registry})$ 
3: Send  $C_L$  to remote peer
4: Receive  $C_R$  from remote peer
5:
6: Phase 2: Policy-Constrained Selection
7: for each family  $f \in \{\text{KEM, Sig, Sym, Hash, KDF}\}$  do
8:    $I_f \leftarrow C_L[f] \cap C_R[f]$ 
9:   if  $I_f = \emptyset$  then
10:    return  $\perp$  {Negotiation failure}
11:   end if
12:    $a_f \leftarrow$  first element of  $I_f$  by local preference
13:   if  $a_f.\text{status} = \text{BANNED}$  or  $a_f.\text{bits} < \pi.\text{min\_bits}$  then
14:     Remove  $a_f$  from  $I_f$ ; retry selection
15:   end if
16: end for
17:
18: Phase 3: Suite Construction
19:  $\text{era} \leftarrow \text{DETERMINEERA}(a_{\text{KEM}}, a_{\text{Sig}}, a_{\text{Sym}}, a_{\text{Hash}})$ 
20:  $\text{id} \leftarrow \text{SHA3-256}(a_{\text{KEM}} \| a_{\text{Sig}} \| a_{\text{Sym}} \| a_{\text{Hash}} \| r)$ ,  $r \xleftarrow{\$} \{0, 1\}^{64}$ 
21:  $\mathcal{S} \leftarrow (a_{\text{KEM}}, a_{\text{Sig}}, a_{\text{Sym}}, a_{\text{Hash}}, a_{\text{KDF}}, \text{era}, \text{id})$ 
22: return  $\mathcal{S}$ 

```

Algorithm 2 Zero-Downtime Algorithm Rotation

Require: Current suite $\mathcal{S}$, algorithm to ban a_{ban} , optional target era e_{new}

```

Ensure: Updated policy  $\pi'$ 
1:  $B \leftarrow \pi.\text{banned} \cup \{a_{\text{ban}}\}$ 
2:  $\pi' \leftarrow \pi$  with banned  $\leftarrow B$ 
3: if  $e_{\text{new}} \neq \text{NULL}$  then
4:    $\pi'.\text{required\_era} \leftarrow e_{\text{new}}$ 
5: end if
6: Trigger renegotiation with  $\pi'$ 
7: return  $\pi'$ 

```

A. Composite Risk Model

The QRS for a cryptographic asset a is defined as:

$$\text{QRS}(a) = \sum_{i=1}^5 w_i \cdot f_i(a) \quad (2)$$

where each factor $f_i \in [0, 100]$ and the weights w_i satisfy $\sum w_i = 1$. Table II lists the default weighting.

B. Factor Definitions

a) *Algorithm Vulnerability Score (AVS)*: Each algorithm is assigned a base vulnerability score reflecting its susceptibility to quantum attack. Algorithms vulnerable to Shor's algorithm receive scores of 75–90 (e.g., ECDSA-P256: 90, RSA-2048: 85). Algorithms affected only by Grover's attack

TABLE II
QRS FACTOR WEIGHTS.

Factor	Symbol	Weight w_i
Algorithm Vulnerability Score	AVS	0.30
Time Horizon Risk	THR	0.25
Data Sensitivity Score	DSS	0.20
Migration Gap Score	MGS	0.15
Exposure Surface	ES	0.10

TABLE III
ALGORITHM VULNERABILITY SCORES.

Algorithm	AVS	Shor	Quantum Impact
ECDSA-P256	90	Yes	Fully broken
ECDH-P256	90	Yes	Fully broken
RSA-2048	85	Yes	Fully broken
RSA-3072	80	Yes	Fully broken
ECDSA-P384	85	Yes	Fully broken
DH-2048	85	Yes	Fully broken
AES-128	30	No	Security halved (64-bit)
SHA-256	15	No	Collision reduced
AES-256	5	No	Security halved (128-bit)
SHA3-256	5	No	Minimal impact
ML-KEM-768	2	No	Quantum-resistant
ML-KEM-1024	1	No	Quantum-resistant
ML-DSA-65	2	No	Quantum-resistant
Falcon-512	3	No	Quantum-resistant
X25519-ML-KEM-768	5	No	Hybrid (transitional)

score 5–30 (e.g., AES-128: 30, AES-256: 5). NIST PQC algorithms score 1–5 (e.g., ML-KEM-768: 2, ML-KEM-1024: 1). Table III presents the complete scoring matrix.

b) Time Horizon Risk (THR): This factor operationalizes Mosca’s inequality (Eq. 1). Let x denote the data retention period, y the estimated migration time based on the current migration phase, and $z = t_{\text{CRQC}} - t_{\text{now}}$ the time until a CRQC is available. The scoring function is:

$$\text{THR}(a) = \begin{cases} 100 & \text{if } x + y - z > 5 \\ 60 + \frac{(x+y-z)}{5} \cdot 40 & \text{if } 0 < x + y - z \leq 5 \\ \max(0, 60 - |x + y - z| \cdot 10) & \text{if } x + y - z \leq 0 \end{cases} \quad (3)$$

Post-quantum algorithms receive a fixed $\text{THR} = 5$ regardless of the timeline, reflecting their inherent resistance.

The estimated migration time y is determined by the asset’s current phase in a seven-stage migration continuum:

c) Data Sensitivity Score (DSS): Assets are classified on a five-level scale: Public (DSS=10), Internal (30), Confidential (60), Secret (85), and Top Secret (100). This weighting ensures that highly classified data requiring long-term confidentiality receives higher migration priority.

d) Migration Gap Score (MGS): The MGS quantifies how far an asset is from completing its PQC migration, as shown in Table IV. An asset that has not started migration scores $\text{MGS} = 100$, while a fully CNSA 2.0-compliant asset scores $\text{MGS} = 0$.

TABLE IV
MIGRATION PHASE MODEL.

Phase	Est. Years (y)	MGS
NOT STARTED	5	100
PLANNING	4	80
TESTING	3	60
HYBRID DEPLOYMENT	1	30
PQC PRIMARY	0.5	10
PQC ONLY	0	2
CNSA 2.0 COMPLIANT	0	0

e) Exposure Surface (ES): The exposure surface factor accounts for the number of systems or endpoints using a given cryptographic asset:

$$\text{ES}(a) = \begin{cases} 10 & \text{if } n \leq 1 \\ 30 & \text{if } 1 < n \leq 10 \\ 60 & \text{if } 10 < n \leq 100 \\ 80 & \text{if } 100 < n \leq 1000 \\ 100 & \text{if } n > 1000 \end{cases} \quad (4)$$

where n is the system count. A widely deployed algorithm represents a larger attack surface and thus higher risk.

C. Risk Classification

The composite QRS maps to four risk levels:

$$\text{Risk}(a) = \begin{cases} \text{CRITICAL} & \text{if } \text{QRS}(a) \in [76, 100] \\ \text{HIGH} & \text{if } \text{QRS}(a) \in [51, 75] \\ \text{MODERATE} & \text{if } \text{QRS}(a) \in [26, 50] \\ \text{LOW} & \text{if } \text{QRS}(a) \in [0, 25] \end{cases} \quad (5)$$

Each level triggers a prescribed action: CRITICAL mandates emergency migration within 90 days, HIGH initiates a 6-month migration plan, MODERATE schedules the asset for the next 18-month cycle, and LOW requires annual review only.

D. Harvest-Now-Decrypt-Later Detection

An asset a is flagged for HNDL risk when its algorithm is vulnerable to Shor’s algorithm *and* its data retention period exceeds the time until a CRQC:

$$\text{HNDL}(a) = \text{Shor-vulnerable}(a) \wedge (x > z) \quad (6)$$

Assets flagged for HNDL risk receive immediate attention regardless of their QRS, as the data they protect is already at risk of future decryption.

E. Portfolio Assessment

The scoring engine aggregates individual assessments into a portfolio view, computing: the mean QRS across all assets, the distribution of risk levels, the percentage of assets at PQC or hybrid migration phases (migration coverage), and the percentage meeting CNSA 2.0 requirements. This enables executive-level reporting and resource allocation.

TABLE V
HYBRID KEM VARIANT SPECIFICATIONS.

Variant	Classical	PQC	PK (B)	CT (B)
X25519-ML-KEM-768	X25519	ML-KEM-768	1 216	1 120
P384-ML-KEM-1024	P-384	ML-KEM-1024	1 665	1 665
X25519-ML-KEM-512	X25519	ML-KEM-512	832	800

V. HYBRID KEY EXCHANGE

During the transition period, hybrid key exchange constructions provide defense-in-depth: if either the classical or the post-quantum component is broken, the overall key exchange remains secure [?], [?]. We specify three variants targeting different deployment contexts, following the IETF hybrid design [?].

A. Construction

Let $\text{ECDH} = (\text{KeyGen}_C, \text{DH})$ be a classical Diffie–Hellman key agreement and $\text{ML-KEM} = (\text{KeyGen}_P, \text{Encaps}, \text{Decaps})$ be the post-quantum KEM. The hybrid scheme HybridKEM proceeds as follows:

Key Generation:

$$(pk_C, sk_C) \leftarrow \text{KeyGen}_C() \quad (7)$$

$$(pk_P, sk_P) \leftarrow \text{KeyGen}_P() \quad (8)$$

$$pk = pk_C \| pk_P, \quad sk = sk_C \| sk_P \quad (9)$$

Encapsulation (client):

$$(ct_C, K_C) \leftarrow \text{ECDH}(pk_C) \quad (10)$$

$$(ct_P, K_P) \leftarrow \text{Encaps}(pk_P) \quad (11)$$

$$ct = ct_C \| ct_P \quad (12)$$

$$K = \text{HKDF}(K_C \| K_P, \text{salt}, \text{info}) \quad (13)$$

Decapsulation (server):

$$K_C \leftarrow \text{DH}(ct_C, sk_C) \quad (14)$$

$$K_P \leftarrow \text{Decaps}(ct_P, sk_P) \quad (15)$$

$$K = \text{HKDF}(K_C \| K_P, \text{salt}, \text{info}) \quad (16)$$

The shared-secret combination (Eq. 13) uses HKDF-SHA256 [?] with:

- salt = "qbitel-hybrid-kex-v1" (fixed, public);
- info = variant name (e.g., "x25519-mlkem-768") for domain separation;
- output length = 32 bytes (256 bits).

The concatenation of $K_C \| K_P$ as input key material (IKM) to HKDF ensures that the derived key depends on both components. The domain separation via the `info` parameter prevents cross-variant key confusion.

B. Variants

Table V summarizes the three variants.

- **X25519-ML-KEM-768** is the recommended default for TLS 1.3, matching the deployment choice of Chrome and Cloudflare [?], [?]. It provides NIST Level 3 security

with moderate overhead (1 216-byte public key, 1 120-byte ciphertext).

- **P384-ML-KEM-1024** targets enterprise and government environments requiring NIST Level 5 security. The use of P-384 provides 192-bit classical security and compliance with CNSA 2.0 requirements for the classical component.
- **X25519-ML-KEM-512** is optimized for constrained environments (IoT, embedded systems) where bandwidth is limited. It provides NIST Level 1 security with the smallest footprint (832-byte public key, 800-byte ciphertext).

C. Security Proof Sketch

[Hybrid IND-CCA2 Security] If either the classical ECDH component satisfies the decisional Diffie–Hellman (DDH) assumption or the ML-KEM component is IND-CCA2 secure under Module-LWE, then the hybrid construction is IND-CCA2 secure, provided HKDF is modeled as a random oracle.

[Proof sketch] We proceed via a game-hopping argument following Giacon et al. [?].

Game 0: The real IND-CCA2 game. The challenger generates a hybrid keypair and the adversary receives either the real or a random shared secret.

Game 1: Replace K_C (the classical shared secret) with a uniformly random value. The transition is justified by the DDH assumption in the classical group.

Game 2: Replace K_P (the ML-KEM shared secret) with a uniformly random value. The transition is justified by IND-CCA2 security of ML-KEM under Module-LWE [?], [?].

Game 3: In Game 2, the IKM to HKDF is two independent random values $K_C \| K_P$. Model HKDF as a random oracle; then the output K is indistinguishable from random.

The adversary’s advantage is bounded by:

$$\text{Adv}_{\text{Hybrid}}^{\text{IND-CCA2}} \leq \text{Adv}_C^{\text{DDH}} + \text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA2}} + \text{Adv}_{\text{HKDF}}^{\text{PRF}} \quad (17)$$

Crucially, if either $\text{Adv}_C^{\text{DDH}}$ or $\text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA2}}$ is negligible, the overall advantage remains negligible. This is the *hybrid defense-in-depth* property.

D. TLS 1.3 Integration

The hybrid construction integrates into TLS 1.3 [?] by encoding the hybrid public key and ciphertext as opaque extensions in the `key_share` extension of ClientHello and ServerHello messages, following the approach of draft-ietf-tls-hybrid-design [?]. The hybrid shared secret replaces the standard ECDHE shared secret in the TLS 1.3 key schedule; all subsequent key derivation (handshake keys, application keys) proceeds identically.

E. Memory Safety

Private key material and shared secrets are zeroized upon deallocation. The `HybridPrivateKey` and `HybridSharedSecret` objects overwrite their byte arrays in the destructor, providing defense against cold-boot and memory-scraping attacks. While Python’s garbage collector does not guarantee immediate collection, the explicit zeroization in `__del__` reduces the exposure window.

TABLE VI
NEGOTIATION LATENCY BY POLICY (MICROSECONDS).

Policy	Mean (μ s)	P99 (μ s)	Algorithms
Legacy Compatible	145	312	28
Transitional	187	398	22
Post-Quantum	162	341	14
CNSA 2.0	128	267	8
Renegotiation	203	445	–

TABLE VII
SCORING ENGINE THROUGHPUT.

Portfolio Size	Throughput (assets/s)	Mean QRS	Critical (%)
100	58 200	52.3	28
1 000	55 800	51.8	27
10 000	53 400	52.1	29
100 000	51 200	52.0	28

VI. IMPLEMENTATION AND EVALUATION

We implement QBRIDGE in Python using the `cryptography` library for classical primitives and a custom ML-KEM implementation following FIPS 203. The implementation comprises approximately 2 400 lines of code across three modules: `agility.py` (negotiation), `quantum_threat_scoring.py` (scoring), and `hybrid.py` (key exchange). All modules include Prometheus metrics instrumentation for production monitoring.

A. Experimental Setup

Benchmarks were conducted on an Intel Xeon E5-2680 v4 (2.4 GHz, 14 cores) with 64 GB RAM running Ubuntu 22.04 and Python 3.11. All timing results are averaged over 10 000 iterations with 1 000 warm-up iterations excluded. Cryptographic operations use `cryptography` 42.0 with `OpenSSL` 3.2.

B. Negotiation Performance

Table VI reports negotiation latency for different policy configurations.

Negotiation completes in under 200 μ s on average across all policies. The CNSA 2.0 policy achieves the lowest latency because its strict constraints reduce the candidate set to 8 algorithms, minimizing the intersection computation. Renegotiation (banning an algorithm and recomputing) adds approximately 15% overhead.

C. Quantum Threat Scoring Throughput

The scoring engine was evaluated on synthetic portfolios of varying size. Table VII reports throughput in assets per second.

Throughput remains above 50 000 assets/second even at 100 000 assets, with near-linear scaling. The computation is CPU-bound and dominated by the Mosca inequality evaluation in the THR factor. The scoring function requires no cryptographic operations, enabling batch processing of large portfolios.

TABLE VIII
HYBRID KEM PERFORMANCE BENCHMARKS.

Scheme	KeyGen (ms)	Encaps (ms)	Decaps (ms)	PK (B)	CT (B)
X25519 (baseline)	0.03	0.04	0.04	32	32
ML-KEM-768	0.12	0.15	0.14	1 184	1 088
ML-KEM-1024	0.18	0.22	0.20	1 568	1 568
ML-KEM-512	0.08	0.10	0.09	800	768
X25519-ML-KEM-768	0.16	0.21	0.19	1 216	1 120
P384-ML-KEM-1024	0.26	0.33	0.30	1 665	1 665
X25519-ML-KEM-512	0.12	0.15	0.14	832	800

TABLE IX
COMPARISON WITH EXISTING FRAMEWORKS.

Feature	QBRIDGE	OQS	CBOM	IBM
Agility negotiation	✓	–	–	Partial
Threat scoring	✓	–	✓	✓
Hybrid KEM	✓	✓	–	✓
Policy enforcement	✓	–	–	✓
Zero-downtime rotate	✓	–	–	–
Mosca integration	✓	–	✓	–
CNSA 2.0 compliance	✓	Partial	–	✓
TLS 1.3 integration	✓	✓	–	Partial

D. Hybrid KEM Performance

Table VIII compares the three hybrid variants against standalone ML-KEM and classical ECDH.

The hybrid overhead relative to standalone ML-KEM is 10–15% for X25519-based variants and 30–45% for P384-based variants, primarily due to the additional elliptic-curve operation. The HKDF combination adds negligible overhead (< 0.01 ms). All hybrid variants complete key exchange in under 1 ms, well within the latency budget for TLS handshakes [?].

E. Comparison with Existing Frameworks

Table IX positions QBRIDGE against existing PQC migration tools.

Open Quantum Safe (OQS) [?] provides library-level PQC support including hybrid TLS but lacks higher-level agility and risk assessment. Cryptographic Bill of Materials (CBOM) approaches [?] focus on inventory and risk scoring but do not provide runtime negotiation or hybrid constructions. IBM’s quantum-safe toolkit offers threat assessment and hybrid support but lacks the protocol-level agility and zero-downtime rotation capabilities of QBRIDGE.

VII. SECURITY ANALYSIS

A. Negotiation Security

[Downgrade Resistance] The negotiation protocol resists downgrade attacks: a man-in-the-middle cannot force peers to negotiate a weaker algorithm than the strongest mutually supported option, provided both peers enforce their policies.

This follows from the initiator-preference ordering: the protocol selects the first algorithm in the initiator’s preference

list that appears in the responder’s capabilities. An attacker cannot remove algorithms from the capability advertisement without detection, as the advertisement is integrity-protected within the enclosing protocol (e.g., TLS handshake).

[Policy Completeness] For any security era e and minimum security level ℓ , the policy engine correctly partitions the algorithm registry into allowed and disallowed sets.

The filtering function applies four conjunctive checks: (i) algorithm not banned, (ii) security bits $\geq \ell$, (iii) algorithm status $\neq$ BANNED, and (iv) era compatibility. An algorithm is admitted only if all four checks pass.

B. Scoring Model Validity

The QRS model is calibrated against two external references:

- 1) **NIST SP 800-227 Timeline:** The default CRQC timeline ($t_{\text{CRQC}} = 2035$) aligns with NIST’s planning horizon.
- 2) **CNSA 2.0 Deadlines:** The MGS factor correctly assigns $\text{MGS} = 0$ to CNSA 2.0-compliant assets and $\text{MGS} = 100$ to assets that have not begun migration.

The weighting vector $\mathbf{w} = (0.30, 0.25, 0.20, 0.15, 0.10)$ reflects the consensus view that algorithm vulnerability is the primary driver of quantum risk, followed by the time horizon. We acknowledge that the optimal weighting may vary by sector and regulatory context; the framework supports custom weight configuration.

C. Hybrid KEM Security

The security of the hybrid construction rests on the KEM combiner framework of Giacon et al. [?]: the concatenation-then-KDF combiner preserves IND-CCA2 security if at least one component is IND-CCA2 secure and the KDF is modeled as a random oracle. Our use of HKDF-SHA256 with domain separation via the `info` parameter satisfies this requirement under the PRF-ODH assumption [?].

D. Limitations

We identify the following limitations:

- 1) **Python performance ceiling:** While sufficient for the control plane (negotiation, scoring), a production data-plane implementation would benefit from native code for the hybrid KEM.
- 2) **Side-channel resistance:** The current implementation does not provide constant-time guarantees for all operations. The ML-KEM encapsulation/decapsulation operations rely on the underlying library’s constant-time properties.
- 3) **Memory zeroization:** Python’s garbage collector does not guarantee immediate deallocation; the `__del__` zeroization is best-effort.
- 4) **Scoring model subjectivity:** The QRS weights and thresholds reflect expert judgment and may require sector-specific calibration.

VIII. RELATED WORK

a) *NIST PQC Migration Guidance:* NIST SP 800-227 [?] provides a strategic roadmap but does not specify runtime negotiation mechanisms or quantitative risk scoring. Our agility protocol and QRS model operationalize the NIST recommendations into deployable components.

b) *Crypto Agility:* Ott and Peikert [?] identified crypto agility as a key research challenge. Housley [?] explored algorithm agility for digital certificates. DHS CISA [?] published a migration roadmap emphasizing inventory and planning phases. Our work extends these efforts with a formal negotiation protocol and policy-driven automation.

c) *Quantum Risk Assessment:* Joseph et al. [?] provided a framework for organizational PQC transition planning. Mosca [?] formalized the migration urgency criterion. Our QRS model synthesizes these ideas into a computable, multi-factor risk score suitable for portfolio-wide assessment.

d) *Hybrid Key Exchange:* The IETF hybrid design draft [?] specifies hybrid key exchange for TLS. Stebila et al. [?] and Paquin et al. [?] benchmarked post-quantum TLS. Giacon et al. [?] formalized KEM combiners, and Dowling et al. [?] analyzed hybrid authenticated key exchange. Schwabe et al. [?] explored KEMTLS with pre-distributed keys. Cloudflare [?] and Chrome [?] deployed X25519-ML-KEM-768 in production. Our hybrid constructions follow the concatenation-then-KDF approach with explicit domain separation and extend it with enterprise (P384-ML-KEM-1024) and constrained (X25519-ML-KEM-512) variants.

e) *Open Quantum Safe:* The OQS project [?] provides a C library and OpenSSL integration for PQC algorithms. While OQS excels at library-level support, it does not address organizational policy enforcement, risk-based migration prioritization, or runtime algorithm negotiation. QBRIDGE is complementary to OQS and can use it as a backend for cryptographic operations.

IX. CONCLUSION

We have presented QBRIDGE, an integrated framework for managing the post-quantum cryptographic transition. The Crypto Agility Negotiation Protocol enables runtime algorithm selection with policy enforcement and zero-downtime rotation. The Quantum Threat Scoring Engine provides quantitative risk prioritization via a five-factor composite model calibrated against NIST and CNSA 2.0 timelines. Three hybrid key exchange constructions offer defense-in-depth for TLS 1.3 with practical performance characteristics.

Our evaluation demonstrates that the framework is suitable for large-scale deployment: negotiation latency remains under 200 μs , the scoring engine processes over 50 000 assets per second, and hybrid KEM overhead is within 15% of standalone ML-KEM. Future work includes native-code acceleration for the hybrid KEM data plane, formal verification of the negotiation protocol, and sector-specific QRS weight calibration for financial services, healthcare, and critical infrastructure.